Griffin Shelor

M.S. Graduate
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Research Interests

Snow Hydrology, Remote Sensing, Geospatial Analysis, Water Resources

Education

University of Nevada, Reno *Reno, NV*

M.S. Geography, August 2025
Thesis: Evaluating the Representativeness of Snow Sampling Sites in the Western United States and Alaska
Advisor: Dr. Anne Nolin

Green Mountain College *Poultney, VT*

B.S. Biology and Natural Resources Management, May 2018

Experience

08/2025-12/2025 **Graduate Research Assistant** *Oklahoma State University, Department of Geography*

Modeled plant functional traits (N, P, K) using hyperspectral imagery in tallgrass prairie ecosystems, evaluated functional diversity using same functional traits, and took meeting notes and scheduled Zoom meetings for the NASA-funded HI-GRASS project.

05/2025-08/2025 **Geospatial Data Analyst** *University of Nevada, Reno*

Analyzed the spatial distribution of the accuracy of machine learning models for predicting precipitation phase using citizen science data collected by the Mountain Rain or Snow project (MRoS) as described in Jennings et al. 2025.

03/2024, 05/2024-09/2024 **Data Analyst and Scientific Programmer** *University of Nevada, Reno*

Used citizen science data collected as part of the Mountain Rain or Snow project (MRoS) to investigate the relationship between probability of atmospheric river data, the Global Precipitation Measurement Mission's (GPM) Probability of Liquid Precipitation data product (PLP), and citizen science observations of precipitation phase in the Cascades and Sierra Nevada EPA Level III Ecoregions, with the primary goal of identifying lower and upper thresholds to use to separate the PLP values into predicted classes of rain, mixed precipitation, and snow.

08/2023-08/2025 **Graduate Teaching Assistant** *University of Nevada, Reno, Department of Geography*

In Spring 2025, I lead lab instruction for GEOG 412/612 (Cartography) which involved supervising lab periods, providing introductions, feedback, and demonstrations during in-class lab periods, as well as grading assignments and troubleshooting technical difficulties with Adobe Illustrator.

In Fall 2024 I lead lab instruction and hosted office hours as the TA for GEOG 416/616 (Spatial Analysis). As part of my duties for this course I provided R coding assistance to students during lab periods, hosted office hours to provide additional technical support, graded lab assignments, and proctored exams.

From Fall 2023-Spring 2024, I lead lab instruction for GEOG 405/605 (GIS I: Geographic Information Systems and Science) which involved preparing lab instructions to ensure they are free of errors and up to date with the latest ArcGIS software, providing introductions, feedback, and demonstrations during in-class lab periods, as well as grading lab assignments and troubleshooting technical difficulties with ArcGIS Pro, ArcGIS Online, ESRI Storymaps, and remote desktop software such as Microsoft Remote Desktop.

10/2022-08/2023 **GIS Team Lead** *University of Vermont Spatial Analysis Lab*

In addition to my GIS Technician 2 and Technician 1 duties, I lead editing operations on a wetlands mapping project for the Lamoille River watershed. I independently designed geoprocessing operations as well as trained new GIS Technicians in the complex methods of wetlands mapping and reviewed their work and provided feedback using National Wetlands Inventory (NWI) guidelines and Vermont Department of Environmental Conservation guidance as a framework. Additionally, I trained other Technicians in general SAL standard

operating procedures, reviewed their work and provided feedback to improve skills and technique, and troubleshooted technical issues and assisted on other SAL projects as needed.

04/2022-10/2022 **GIS Technician II** *University of Vermont Spatial Analysis Lab*

Assisted with higher-level projects mapping land cover and multi-temporal tree canopy change. Reviewed the work of newer SAL GIS Technicians and provided feedback to improve skills and technique, and troubleshooted technical issues and assisted on other SAL projects as needed. Continued duties from GIS Technician I position.

04/2021-04/2022 **GIS Technician I** *University of Vermont Spatial Analysis Lab*

Digitized land cover and tree canopy change features using NAIP imagery, aerial orthoimagery, and LiDAR. Performed basic geoprocessing, edited spatial databases, and created maps visualizing land cover and tree canopy change. Edited LiDAR Digital Elevation Models (DEMs) to create hydro-enforced DEMs for a flood inundation modeling research project.

12/2018-01/2021 **Cartographic Technician** *U.S. Census Bureau*

Utilized ArcGIS software and Census Bureau GIS software programs to analyze and solve complex problems including correction of legal boundaries, consulting with county and municipality contacts, digitizing geographic updates based on satellite imagery, resolving data pertaining to public land boundaries, and geocoding 2020 Decennial Census responses.

06/2018-08/2018 **Remote Sensing Intern** *NASA DEVELOP National Program*

Conducted a supervised land cover classification in ArcMap 10.5. Used Random Forest modeling to identify important predictor variables for a separate Random Forest model in Google Earth Engine (GEE). Created fire masks to better identify possible bison-related disturbance. Assisted with the creation of several deliverables including a public-facing website image, a poster, a Powerpoint presentation, a technical report, as well as a documentation package with our data, other deliverables, and a README to be handed off to our project partners at USGS.

Research and Publications

Technical Reports

[1] LaRoe, J., Rasmussen, S., Schee, M., **Shelor, G.**, *Employing Landsat to Model Availability of Ephemeral Water Sources and Vegetation Change in Support of a USGS Feasibility Assessment and Management Strategy of Bison*, NASA DEVELOP National Program, 2018

Preprints

[1] **Shelor, G.**, Nolin, A., *Evaluating the Representativeness of Snow Sampling Sites in the Western United States and Alaska*, (*In Submission*), 2026

Poster

[1] **Shelor, G.**, Nolin, A., *Using SHAP to Derive Peak SWE-based Snow Cover Classes To Analyze Sampling Site Representativeness*, Western Snow Conference, Bozeman, MT, 2025

[2] **Shelor, G.**, Van Hoesen, J., *A LIDAR-Based Approach to Derivative Map Production and Visualizing Bedrock Topography Within The Towns of Proctor and Pittsford, Vermont, USA*, Northeast Arc Users Group Conference, Storrs, CT, 2018

Grants and Funding

2025 **UNR Graduate Student Association Travel Award**. Role: Presenting at the Western Snow Conference *$500*

Academic Service

2017-2018 Co-President and Treasurer, Green Mountain College Student Chapter of The Wildlife Society *Green Mountain College*

2014, 2016, 2017 Vermont Department of Fish and Wildlife Deer Check Station Volunteer *Green Mountain College*

Teaching and Mentoring Experience

Teaching

2025 **Graduate Teaching Assistant**, GEOG 412/612: Cartography *University of Nevada, Reno*

2024 **Graduate Teaching Assistant**, GEOG 416/616 Spatial Analysis *University of Nevada, Reno*

2023-2024 **Graduate Teaching Assistant**, GEOG 405/605 GIS 1: Geographic Information Systems and Science *University of Nevada, Reno*

Awards and Honors

05/2018 Eagle Award *Green Mountain College*

05/2018 Excellence in Mathematics Award *Green Mountain College*

05/2018 Natural Resources Management Senior Award *Green Mountain College*

05/2018 Scholar-Athlete Award *Green Mountain College*

05/2017 Scholar-Athlete Award *Green Mountain College*

Skills

Programming Languages R, Python, Stan

Software/Tools RStudio, Positron, ArcGIS Pro/Online, ArcPy, ESRI Storymaps, QGIS, Google Earth Engine (Python/Javascript), Microsoft Planetary Computer, ENVI, Git/Github, OpenDroneMap, Pix4DMapper, Microsoft Office, LaTeX, Typst

Professional Memberships

American Association of Geographers 2018 - Present

American Geophysical Union 2022 - Present

Relevant Coursework

- **UNR**
- GEOG 607: Advanced GIS Analysis
- GEOG 611: Remote Sensing: Principles and Applications
- GEOG 613: UAS Technologies in Geography
- GEOG 616: Spatial Analysis
- GEOL 679: Python for Earth Sciences
- NRES 776: Ecological Forecasting
- NRES 779: Bayesian Hierarchical Modeling

Additional Research Experience

09/2021-01/2022 **Creating Better Flood Inundation Maps with LiDAR Data and a Topographically-driven Model** *University of Vermont Spatial Analysis Lab*

I worked with a team of fellow SAL GIS Technicians to use ArcGIS Pro to manually hydro-enforce LiDAR DEMs so that they properly reflected the topography of the landscape in service of a project led by UVM's Dr. Rebecca Diehl and Stephanie Drago to increase the accuracy of flood inundation modeling. Link to Presentation

01/2018-05/2018 **Mapping Canine parvovirus in Mesocarnivores in Vermont** *Green Mountain College*

I created cartographic products used to depict where mammal scat samples were found, as well as which scat samples tested positively for *Canine parvovirus* (CPV). I additionally performed a geostatistical analysis of spatial autocorrelation of CPV throughout the southwestern region of Vermont.

References

1. **Dr. Anne Nolin**
   Professor, Department of Geography
   University of Nevada, Reno
   Email: anolin@unr.edu
   Phone: +1-775-682-8148

2. **Dr. Kenneth Nussear**
   Associate Professor, Department of Geography
   University of Nevada, Reno
   Email: knussear@unr.edu
   Phone: +1-775-784-6612

3. **Dr. Scott Kelley**
   Associate Professor, Department of Geography
   University of Nevada, Reno
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4. **Anna Royar**
   GIS Technical Lead
   University of Vermont Spatial Analysis Lab
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